Backtesting Probability of Default Models

Adjusting the Z-Score Test for Portfolio Heterogeneity

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Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to adhering to the prescribed risk quantification process, practitioners are required to assess the appropriateness of model estimates. This is typically referred to as backtesting.
- The primary purpose of backtesting in this context is to ensure that model estimates accurately predict the occurrence of defaults, i.e., that PD estimates provide reliable forecasts of default rates.
- In practice, this is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- Regardless of the type of PD model output (discrete or continuous), the model should be backtested across the full range of estimates and at the portfolio and segment levels.
- When moving from rating-grade to portfolio-level testing for discrete model outputs, practitioners often overlook that portfolio (segment or subsegment) data consist of heterogeneous obligors assigned estimates derived from homogeneous rating grades.

Probability of Default Backtesting cont.

- Practitioners typically assume that, at the portfolio level, PD follows a binomial distribution, where PD is the average probability of default across all obligors and the corresponding variance is given by $\frac{PD(1-PD)}{N}$.
- However, given that the primary level of estimation for discrete model outputs is the rating grade, it is reasonable to assume that portfolio-level defaults arise from a series of independent Bernoulli trials that are not necessarily identically distributed. Moreover, considering the requlatory requirement for rating scale heterogeneity, it is evident that rating-grade PDs are not identically distributed.
- Overlooking this fact leads to the inappropriate application of certain statistical tests for the purpose of PD backtesting at the portfolio or subsegment level.
- The Poisson binomial distribution describes the discrete probability distribution of the sum of independent Bernoulli trials that are not necessarily identically distributed.
- This presentation demonstrates an adjustment to the standard z-score test for backtesting that accounts for PD heterogeneity at the portfolio or sub-segment level. The following slides introduce the Poisson binomial distribution, present the standard and portfolio-adjusted z-score statistics, and compare their Type I error rates through a simulation study.

Poisson binomial distribution

- The Poisson binomial distribution describes the discrete probability distribution of the sum of independent Bernoulli trials that are not necessarily identically distributed.
- Unlike the binomial distribution, which assumes homogeneity of the success probability across observations, the Poisson binomial distribution accounts for heterogeneity in the probabilities assigned to the analyzed observations.
- For the purpose of PD backtesting, the concept of the Poisson binomial distribution naturally applies to portfolio-level or subsegment-level testing, given that the output of PD models is typically calibrated at the level of homogeneous rating grades.

Poisson binomial distribution cont.

The following formulas define the main characteristics of the Poisson binomial distribution:

1. Mean:

$$\mu = \mathbb{E}[X] = \sum_{i=1}^{n} p_i$$

2. Variance:

$$\sigma^2 = \mathrm{Var}(X) = \sum_{i=1}^{n} p_i(1 - p_i)$$

3. Probability Mass Function(PMF):

$$P(X = k) = \sum_{A \in F_k} \prod_{i \in A} p_i \prod_{j \in A^c} (1 - p_j), \qquad k = 0, 1, \ldots, n$$

where F_k, denotes the collection of all subsets of $1, \ldots, n$ containing exactly k elements and A^c is the complement of A.

Given the above characteristics, the Poisson binomial distribution can be used as a natural extension of the standard binomial test for PD backtesting, while properly accounting for heterogeneity among rating-grade PD estimates.

The Standard Z-Score Test

The test statistic of the z-score test is given as follows:

$$z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{N}}}$$

where:

- ODR is the observed default rate;
- PD is the average calibrated PD;
- N is the number of exposures.

Under the assumption that the z_{score} test statistic follows a standard normal distribution, a `p-value` is calculated accordingly.

The Portfolio-Adjusted Z-Score Test

The test statistic of the portfolio-adjusted z-score test is given as follows:

$$z_{score}^{pb} = \frac{ODR - PD}{\frac{\sqrt{\sum_{i=1}^{G} n_i PD_i(1-PD_i)}}{N}}$$

where:

- ODR is the observed default rate;
- PD is the calibrated PD at the portfolio level;
- PD_i is the calibrated PD at rating grade i;
- n_i is the number of exposures in rating grade i;
- N is the number of exposures in the portfolio;
- G is the number of rating grades.

Under the assumption that the z_{score}^{pb} test statistic follows a standard normal distribution, a `p-value` is calculated accordingly.

Simulation Study

The purpose of this simulation study is to assess the statistical validity of the portfolio-adjusted z-score test by evaluating its Type I error rate at different significance levels. Specifically, the simulation study compares the p-value distribution and Type I error rate of the z-test under the assumptions of homogeneity (standard z-score test) and heterogeneity (portfolio-adjusted z-score test) of obligors at the portfolio level.

The simulation is based on the following rating scale:

```
##   rg    n    pd
##  RG1  300 0.005
##  RG2 1000 0.010
##  RG3 2200 0.025
##  RG4 3000 0.040
##  RG5 1800 0.070
##  RG6 1200 0.200
##  RG7  500 0.350
```

Simulation Study cont.

The following steps outline the simulation design:

1. for a given rating scale, calibrated PD, and number of obligors, independently simulate the number of defaults for each rating grade from a binomial distribution, with the probability equal to the calibrated PD and the sample size equal to the number of obligors;
2. based on the simulated number of defaults, calculate the default rate for each rating grade;
3. calculate the p-value of the standard z-score test applied at the portfolio level to test for evidence of PD underestimation (one-sided test);
4. calculate the p-value of the adjusted z-score test applied at the portfolio level to test for evidence of PD underestimation (one-sided test);
5. repeat steps 1 to 4 10,000 times and store the simulated p-values;
6. for both tests, calculate the Type I error rate as the percentage of simulations that reject the null hypothesis at significance levels of 1%, 5%, and 10%;
7. plot the p-value distributions for both tests.

Simulation Results

Type I error:

```
##                                  alpha = 1% alpha = 5% alpha = 10%
## Naive Z-Score Test               0.70%      3.92%      8.65%
## Portfolio-Adjusted Z-Score Test  0.99%      5.08%      10.15%
```

